

Before I Wake: A First-Person 3D Escape Room Puzzle Game

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April 26

1 Abstract

This report presents the design and implementation of *Before I Wake*, a first-person 3D escape-room puzzle game prototype developed in Unity. The game is set in a night-locked school building and combines light horror, environmental storytelling, and logical puzzle-solving. The player wakes up in a dark classroom and explores four connected rooms: the classroom, reading room, chemistry lab, and principal's office. Each room contains a different puzzle based on observation, classification, material selection, or associative reasoning. After solving the puzzles, the player obtains four digits and rearranges them according to the clue shown on the password door to escape. The game uses simple first-person controls, a dual-mode interaction system, visual and audio feedback, and UI hints to support player understanding. Development was carried out using Unity and selected Unity Asset Store resources. Preliminary playtesting showed that all test players completed the game, with an average completion time of approximately 13 minutes, suggesting that the prototype is playable and has a manageable difficulty level. Although some interactions remain simplified, the project successfully demonstrates a complete gameplay loop and a coherent dreamlike escape-room experience.

2 Keywords

First-person game; Escape room; Puzzle design; Environmental storytelling

3 Introduction

The goal of this project is to develop a first-person 3D escape room/puzzle game titled *Before I Wake*. Set on a school campus, the game blends darkness, a dreamlike atmosphere, and light horror. The player wakes up in a pitch-black classroom and must explore four rooms connected by a corridor, solve each room's unique puzzle to obtain four digits, and then rearrange them according to the order shown on the password door to escape.

The unique highlights of the game are:

1. **Narrative and gameplay integration:** The ending reveals that “everything was just a dream”, enhancing immersion and emotional impact.
2. **Progressive tutorial:** The classroom serves as a tutorial, teaching pickup, item switching, and light interaction; later rooms increase puzzle complexity.
3. **Diverse puzzle types:** The game includes UV development, book classification, chemical reagent reactions, and identity inference, covering observation, classification, experimentation, and searching.
4. **Atmospheric environment:** Avoiding jump scares, the game builds an oppressive and dreamlike feeling through dim lighting, cool colors, ambient sounds, and slightly disordered object placement.

This game draws on the environmental narrative and psychological horror of *Layers of Fear* [1], and the multi-room puzzle structure of *Escape Simulator* [2].

4 Game Design

4.1 Game Concept

The core concept of *Before I Wake* is “logical escape within a dream”. The player wakes in an oppressive school, solves four similar but independent puzzles through exploration and reasoning, and then rearranges the obtained digits to escape. The game combines light horror with puzzle-solving achievement, while the ending twist, where everything is revealed to be a dream, strengthens the integration between narrative and gameplay. The design keeps players tense but not terrified, allowing clear logical thinking.

4.2 Game World

The game takes place in a night-locked school building. The room distribution is as follows:

- **First floor:** Classroom (A, spawn point), Chemistry Lab (C), Principal’s Office (D), and the exit.
- **Second floor:** Reading Room (B).

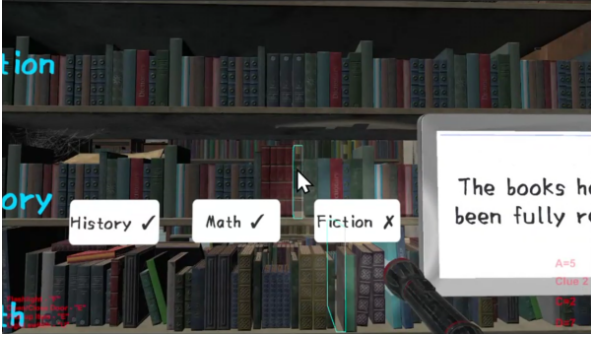
A corridor connects the areas. The Principal’s Office (D) is inside Room A, away from the blackboard. The password door, which serves as the exit, is behind the staircase connecting the two floors. Other rooms, including Room A, Room C, and the stairs, are accessed from the corridor. The design rationale is as follows:

- **Vertical layering and hidden room:** The Reading Room on the second floor adds depth, while the hidden office inside the classroom breaks conventional expectations.
- **Semi-free exploration under a linear skeleton:** Starting in Room A, players must get basic lighting before exploring Room B, Room C, or Room D. After that, any order is allowed, ensuring a tutorial while avoiding full linearity.
- **Atmospheric environmental storytelling:** Darkness, cool colors, limited light from the flashlight, ambient sounds, and slightly messy objects create a dreamlike feeling without relying on jump scares.
- **Functional zoning and thematic consistency:** Each room’s function, such as teaching, reading, laboratory work, or administration, directly suggests its puzzle theme, reducing cognitive load.

4.3 Game Interaction

The interaction system follows the principle of “simple, direct, and minimal interruption”.

- **First-person view:** The game uses a first-person view with a reticle. The detection area is generous, so precise aiming is not required.
- **Dual-mode interaction:**
 - **Key-based interaction:** High-frequency actions, such as picking up items and opening doors, are performed without calling up the mouse cursor.
 - **Mouse-cursor interaction:** Complex interactions, such as placing books, pouring reagents, using UV light, and entering computer passwords, are performed by pressing Q. This freezes the view and brings up the cursor, as shown in Fig. 1(a). Pressing Q again exits this mode. This makes fine selections easier, at the cost of a brief interruption.
- **Interaction feedback:** Successful actions trigger sound or visual feedback, while incorrect actions play an error sound but cause no penalty. Players can retry freely without progress reset.
- **UI and prompts:** The bottom-left UI displays key hints, such as “Pick up – E” and “Light switch – U”. The bottom-right UI displays dynamic clues, such as “A = 5”. Progressive hints, as shown in Fig. 1(b), appear near key objects to narrow the search without directly giving answers.



(a) Mouse-cursor interaction



(b) Progressive hint example

Figure 1: Examples of interaction and hint design in the development environment.

4.4 Gameplay Mechanics and Theory

The core mechanic is “distributed puzzles + code reassembly”. Each room yields one digit. Unlike many escape-room games, the order is not A–B–C–D, but BCDA, which is shown on the password door. After solving all rooms, the UI shows $A = 5$, $B = 8$, $C = 2$, $D = 7$; therefore, the correct password is 8275. This design uses the bottom-right UI to support memory, allowing players to focus on replacing each letter with its corresponding digit. It also breaks the linear A–B–C–D expectation and uses a simple cyclic shift to reduce frustration.

4.5 Ending Design

After entering 8275, the password door opens to a soft white glow. As the player steps out, the screen over-exposes and cuts to a bright, normal classroom with background school break noise, such as chatter, footsteps, and moving desks. Fig.2(a) shows the classroom scene after the ending. This reveals that the whole experience was a dream. The environmental contrast enhances the emotional return to reality and provides psychological relief.

4.6 Puzzle Type Design

The four rooms correspond to different cognitive skills, forming a progressive difficulty curve. Fig.2(b) shows the classroom scene after the ending and summarises the puzzle types used in each room.



(a) Classroom scene after the ending.

Room	Puzzle Type	Core Skill	Purpose
A	Tutorial + UV	Tool use, observation	Learn basic interactions
B	Classification & placement	Classification, matching	Test environmental understanding
C	Material selection	Reading inference, choice	Simulate experiment logic
D	Search & associative reasoning	Searching, information linking	Connect clues to numeric codes

(b) Puzzle type design.

Figure 2: Classroom scene and puzzle type design.

4.7 Puzzle Design Details

1. **Classroom (A) – UV development:** After acquiring the flashlight and UV light, using F to toggle the flashlight and U to switch mode, players can shine the UV light on the blackboard and find the hidden equation $3 + 2 = 5$. This simple puzzle builds confidence and teaches flashlight use.

2. **Reading Room (B) – Classification and placement:** An electronic screen shows that the books have not been fully returned. Categories and gaps on the shelves indicate the required books. Players aim the reticle at a shelf gap and press **E** to place a book. Correct placement triggers sound feedback, while wrong placement plays an error sound without penalty.
3. **Chemistry Lab (C) – Simplified material selection:** The player reads the handbook hint: “Blue ink + oxalic acid $\rightarrow$ ink transparent”. By pressing **Q**, the player brings up the cursor and selects the correct oxalic acid bottle. The system automatically triggers the pouring effect, the ink fades, and clue “2” appears. This retains the logic of reading instructions and selecting the correct material.
4. **Principal’s Office (D) – Search and inference:** The player finds the hidden entrance inside Classroom A. Inside the office, the player searches the coat pocket for the principal’s ID card and reads the birthday. By pressing **Q**, clicking the computer, and entering the birthday as the password, the desktop is unlocked and shows “Lucky number 7”. This puzzle emphasises hidden clues and common-sense reasoning.

4.8 Atmosphere and Rhythm Control

The game’s rhythm is designed as “tension–relief–tension”. Corridor exploration is free and quiet, providing relief, while room puzzle solving requires focused attention and creates tension. Each room puzzle is designed to take around 2–5 minutes, and the total gameplay time is approximately 15–25 minutes, which is suitable for a single-session escape-room experience.

5 Implementation

5.1 Division of Work

Table 1 summarises the main responsibilities of each team member. Although each member had a primary role, all team members contributed to game design, code modification, testing, and the preparation of the final report.

Name (Student ID)	Responsibility
Jiahui Du (2252588)	Responsible for overall gameplay design, task allocation, and progress management. Designed the puzzle flow and story background.
Zhebing Xia (2252461)	Implemented player movement and item interaction using Unity and C#.
Wendi Shen (2252248)	Developed puzzle mechanisms such as password locks, triggers, and event logic.
Lili Tao (2251893)	Built environments, arranged models, and handled lighting and visual effects.
Ziyun Gu (2255736)	Designed sound effects and narrative content, conducted playtesting, and documented player experience feedback.

Table 1: Division of work among team members.

5.2 Development Process

The development process followed a structured workflow. First, the team designed the overall game concept, including the story background, level structure, and puzzle logic. Next, the game environment was constructed in Unity, where different rooms and scenes were created and arranged. After that, core gameplay systems such as player movement, object interaction, and puzzle mechanisms were implemented using C#. Then, user interface elements and interaction systems were integrated to support gameplay actions such as item usage and password input. Finally, the game was tested and refined through multiple iterations to improve stability and player experience.

Specifically, these systems were implemented using Unity components such as colliders, input detection, and UI elements, with C# scripts controlling interaction logic and state updates. The interaction system combines trigger detection and player input. Trigger colliders detect when

the player enters specific areas and display contextual hints, which are hidden upon exit. For object interaction, the system checks both player distance and input. When within range and the interaction key is pressed, actions such as opening a password panel are triggered. Puzzle progression is managed using state variables. For example, in the password system, the player’s input is validated against a predefined value. A correct input updates the puzzle state, reveals new clues, and unlocks content, while incorrect input provides feedback and prompts retry. UI-based interaction is implemented using input fields and text components, which update dynamically to clearly communicate feedback and puzzle information.

For example, in the Chemistry Lab puzzle, interaction with reagent bottles is used to reveal hidden clues. When the player clicks a reagent bottle, the interaction triggers a `DOTweenAnimation` component, which is supported by the DOTween plugin, a lightweight tweening tool integrated into Unity [3]. This rotates the bottle to simulate a pouring motion. At the same time, an audio source is activated to provide a pouring sound effect, enhancing the realism of the interaction. After the animation is completed, the system updates the puzzle state by recording the corresponding clue value. This is implemented using a state variable that changes when the interaction is finished. As a result, the UI is updated to display a new clue, such as “C = 2”, replacing the previous placeholder and providing clear feedback to the player. This process demonstrates how player input, animation effects, state updates, and UI feedback are integrated to support puzzle progression.

5.3 Tools and Technologies

The project was developed using Unity 2020.3.48 as the main game engine and C# for scripting and system implementation. Visual Studio was used for coding and debugging. Most visual assets, including the indoor and outdoor scene structures and thematic environment elements, were obtained from the Unity Asset Store package *HQ Abandoned School (Modular)* [4]. These assets were used to construct the main game environment and establish the abandoned school atmosphere. ChatGPT was used as a supporting tool for debugging, code understanding, and report writing.

5.4 Development Challenges

During the development process, one major challenge was designing gameplay that was both engaging and intuitive. Since the game includes multiple types of puzzles, such as classification, experimentation, and logical reasoning, each puzzle needed to be understandable while still maintaining a suitable level of difficulty. To address this, the team incorporated a progressive tutorial in the classroom and used highlighted text and bounding boxes to guide the player, as shown in Fig.3.

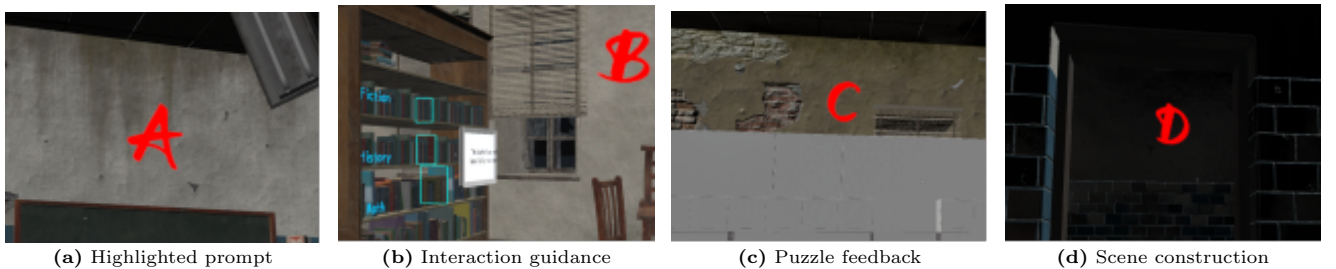


Figure 3: Examples of player guidance, interaction feedback, and scene implementation.

Another challenge was the construction of complex scenes and the selection of appropriate assets. Due to the detailed environment and the need to create a consistent abandoned school atmosphere, finding suitable assets and organising them efficiently was time-consuming. To solve this, the team used Unity Asset Store resources and carefully adjusted model placement, lighting, and post-processing effects to achieve visual consistency.

5.5 Preliminary Results and Player Experience

Initial playtesting results showed that the game provides smooth controls and responsive interactions. Players were able to understand the gameplay and successfully complete all puzzles with guidance. Table 2 summarises the playtesting results.

Player	Completion Time (min)	Completed	Difficulty (1–5)	Most Difficult Puzzle
P1	12	Yes	2	Chemistry Lab
P2	10	Yes	1	Reading Room
P3	14	Yes	2	Chemistry Lab
P4	16	Yes	4	Classroom

Table 2: Playtesting results.

Based on the collected data, the average completion time was approximately 13 minutes. All players were able to complete the game, indicating that the overall difficulty is manageable. The average difficulty rating was around 2–3 out of 5, suggesting an easy to moderate level of challenge. Among the puzzles, the Chemistry Lab was most frequently reported as the most difficult, indicating that it may require further balancing or clearer guidance.

Further player feedback highlights puzzle difficulty differences. As the classroom is the initial scene, some players experienced confusion while learning the game mechanics, particularly due to the non-intuitive use of the UV light.

The chemistry laboratory was most frequently identified as the most difficult puzzle, with players uncertain about the required number of reagent bottles, suggesting the need for clearer guidance.

6 User Manual

6.1 Getting Started

- Extract the ZIP folder contents.
- Double-click `Group_8_A2_ExportedExecutiveFile.exe` to launch the game and access the main menu.
- Wait a few moments for the game to load based on your system performance.

6.2 System Requirements and Controls Overview

The game requires specific system configurations for optimal performance. The basic system requirements and controls are shown in Table 3.

Component	Requirement	Key / Input	Function
Operating System	Windows 10 or later	W, A, S, D	Move the character
Processor	Intel Core i5 or equivalent	Mouse	Look around and control the view-point
Memory	8 GB RAM	Space	Jump
GPU	DX10 compatible graphics card, Shader Model 4.0	Shift	Sprint
Storage	5 GB available space	E	Pick up items and open or close doors
		F	Turn the flashlight on or off
		U	Turn the UV light on or off
		Left Click	Interact with text boxes, shelves, and objects
		Q	Lock the camera for easier puzzle interaction
		Tab	Open the bag

(a) System requirements.

(b) Controls overview.

Table 3: System requirements and controls overview.

6.3 Gameplay Instructions

6.3.1 Game Structure

The game takes place on a school floor with multiple rooms, as shown in the Player Exploration Map. Each room contains unique puzzles and hidden clues that aid in solving them. Players should interact with objects in their surroundings carefully, as many puzzles may require trial and error to progress.

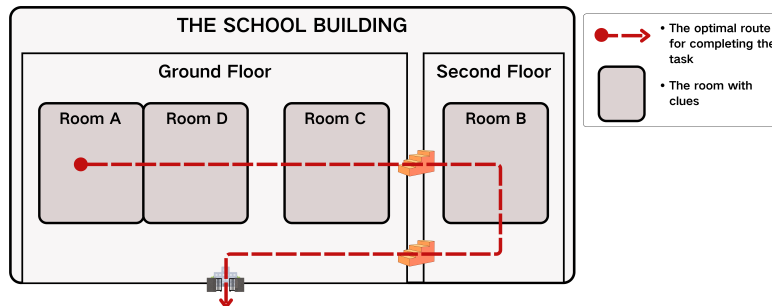


Figure 4: Player exploration map.

6.3.2 Classroom (Room A)

Upon entering the classroom, the player will find a flashlight on the desk. Press **E** to pick it up. Use **F** to turn on the flashlight, and press **U** to switch it to UV light mode. The player can then use the UV light to reveal the first clue hidden on the blackboard. This gives Clue A.

Once Clue A is collected, the player can head towards the back of the classroom and find the door to the Principal's Office.

6.3.3 Principal's Office (Room D)

Upon entering the Principal's Office, the player will notice a computer and a piece of clothing. First, the player can try interacting with the computer and entering a random password, but it will fail and give a hint that the password is related to the principal.

Next, the player should interact with the clothing by pressing **E**. The player can pick up the principal's ID card, which reveals the principal's birthday. With this information, the player can return to the computer and input the birthday as the password to gain access to Clue D.

The player should press **Q** to lock the camera before interacting with the computer, so that it is easier to click on the input box and type the password.

6.3.4 Chemistry Lab (Room C)

In the Chemistry Lab, the player will find three bottles of chemicals and a beaker with purple liquid. The goal is to make the liquid in the beaker turn transparent according to the hints from the book beside it.

The player should press **Q** to lock the camera, and then use the chemicals from the bottles to mix the solution in the beaker. Once the correct chemical is selected, the liquid will turn transparent and reveal Clue C. After collecting Clue C, the player can exit the Chemistry Lab and proceed to the next floor.

6.3.5 Reading Room (Room B)

Upon reaching the Reading Room on the second floor, the player will find a bookshelf with missing books. The player needs to locate and collect three books: Math, History, and Fiction.

The Fiction book is located on the window sill, and the player needs to jump onto the ledge to reach it. After collecting all three books, the player should return to the bookshelf and press **Q** and **E** to interact with it. The mouse can then be used to place the books in the correct order. Once completed, the player will receive Clue B.

6.3.6 Password Door and Escape

Once all clues, Clue A, Clue B, Clue C, and Clue D, are collected, the player should return to the first floor and reach the exit door. The final password is in the order of Clue B, Clue C, Clue D, and Clue A, which forms the code 8275. After inputting the password, the player escapes from the school and completes the game.

Upon escaping, the player returns to the classroom and realises that the whole experience was a dream.

7 Discussion

Before I Wake successfully demonstrates the core experience of a first-person 3D escape-room prototype. Its main strength is the integration of narrative, environment, and puzzle design. The abandoned school setting, dark lighting, flashlight-based exploration, and final dream reveal all support the concept of “logical escape within a dream”. Instead of relying on jump scares, the game creates tension through limited visibility, ambient sound, and environmental uncertainty, allowing players to stay focused on logical puzzle-solving.

The game also provides varied puzzle types across the four rooms, including observation, classification, material selection, and associative reasoning. The classroom puzzle works as a simple tutorial, while later rooms introduce more complex tasks such as book placement, chemical selection, and password inference. Playtesting suggests that the game is playable and understandable, as all four test players completed the game with an average completion time of approximately 13 minutes. However, some mechanics are still simplified due to development time and technical constraints.

Several ethical and practical issues were considered. Since the game uses a dark school environment and light horror atmosphere, it should avoid excessive jump scares or disturbing content, and a short content notice could be added before the game starts. External assets from the Unity Asset Store and the use of ChatGPT as a supporting tool should also be clearly acknowledged. Future work could improve puzzle balance, interaction depth, accessibility options, sound effects, and animation. If further developed for VR, physical interactions such as picking up objects, placing books, and pouring reagents could make the experience more immersive.

8 Conclusion

In conclusion, *Before I Wake* achieves the goal of creating a functional first-person 3D escape-room game prototype. The project includes a complete gameplay loop, a coherent dreamlike school setting, multiple puzzle types, basic UI support, player guidance, and a playable ending. Through exploration, clue collection, puzzle-solving, and final code reassembly, the game presents the intended experience of “logical escape within a dream”.

The project also shows the team’s ability to connect game design concepts with practical Unity implementation. Atmospheric lighting, sound feedback, progressive hints, and room-based puzzles help create an understandable and engaging player experience. Although some mechanics remain simplified, the prototype effectively communicates its core design idea and provides a solid foundation for future development, such as deeper interactions, improved puzzle balance, enhanced audiovisual polish, and possible VR adaptation.

References

- [1] Bloober Team. *Layers of Fear*. Bloober Team, 2016.
- [2] Pine Studio. *Escape Simulator*. Pine Studio, 2021.
- [3] Demigiant. *DOTween Documentation*. Available at: <http://dotween.demigiant.com/documentation.php>
- [4] Unity Asset Store. *HQ Abandoned School (Modular)*. Unity Technologies.